Applying this to real 5-minute data for the GBL and GBM EUREX futures contracts, following Kakoullis (2010), first rebase the data to aid visual inspection and test for stationarity using ADF tests. The rebased price series alongside the rebased spread series can be seen in Figure 13.2(a) and Figure 13.2(b) respectively. Tables 13.1 and 13.2 indicate we can reject stationarity at the 5% significance level (as the spread t-statistic $>$ 5% critical significance level). It can thus be concluded that both GBL and GBM are non-stationary.

Secondly, test for cointegration by running an Engle-Granger regression of the form

$$GBL_t^{reb} = a + bGBM_t^{reb} + \varepsilon_t \tag{13.11}$$

and test the error variable term ε_t for stationarity. Following the same t-statistic test as above it can be said that the spread is weakly stationary, thus GBL and GBM are stationary with cointegrating vector $(1, -b)$. Thirdly, build an ECM of the form of Equation 13.4. A VECM is built using a general-to-specific approach: a general model is initialized with $i = 5$ (five period lags) for both series which are then used to systematically remove insignificant variables and test the error variable term ε_t for stationarity. Following the same t-statistic test as above it can be said that the spread is weakly stationary.

Finally, as the coefficients of the disequilibrium term W_{t-1} have opposite signs and both are highly significant given their t-statistic, then an error correction mechanism from long-term deviations exists. Furthermore, the magnitude of the coefficients gives an indication of the timing until adjustment, e.g., $\gamma 1 = -0.027$ can be interpreted as a 2.7% adjustment in 5 minutes or 3 hours for mean-reversion. Corollary, as the coefficient of $\Delta GBM^{reb}(-1)$ is significant in Equation 13.11 we can say that GBM leads GBL.

Triangular Arbitrage

Triangular arbitrage exploits mispricing across at least three foreign exchange (FX) rates. Consider the scenario where you initially hold x_i dollars. If you sell these dollars to buy euros, convert these euros to pounds, and finally convert these pounds into x_f dollars then you will realize a profit if $x_f > x_i$. If the intermediate rate does not exist you can calculate the synthetic cross to complete the exchange.

In highly liquid markets, such opportunities should be limited and if they were to exist you would expect to find the difference $x_f - x_i$ to be very small (Daniel et al., 2009). Given this restriction, when such